MODULE – 2

SENSORS

A sensor is a device that measures physical input from its environment and converts it into data that can be interpreted by either a human or a machine.

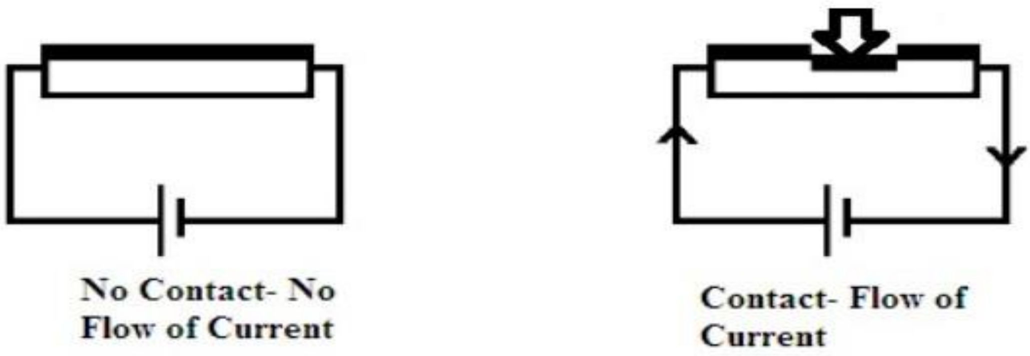
Sensor Classification

Touch sensor

Definition: A sensor that is capable enough of capturing and recording the physical touch made by the operator can be defined as a touch sensor. Further, it is also referred to as a touch detector.

Touch Sensor Working Principle

These sensors are sensitive to any pressure or force applied or touch. The principle of touch is similar to that of a switch. When the switch is closed, the current flows otherwise there is no chance of the current to flow. Similarly, when the touch sensor senses the touch or proximity is captured then it acts like a closed switch otherwise it acts as an open switch. These sensors are also known as 'Tactile Sensors'.

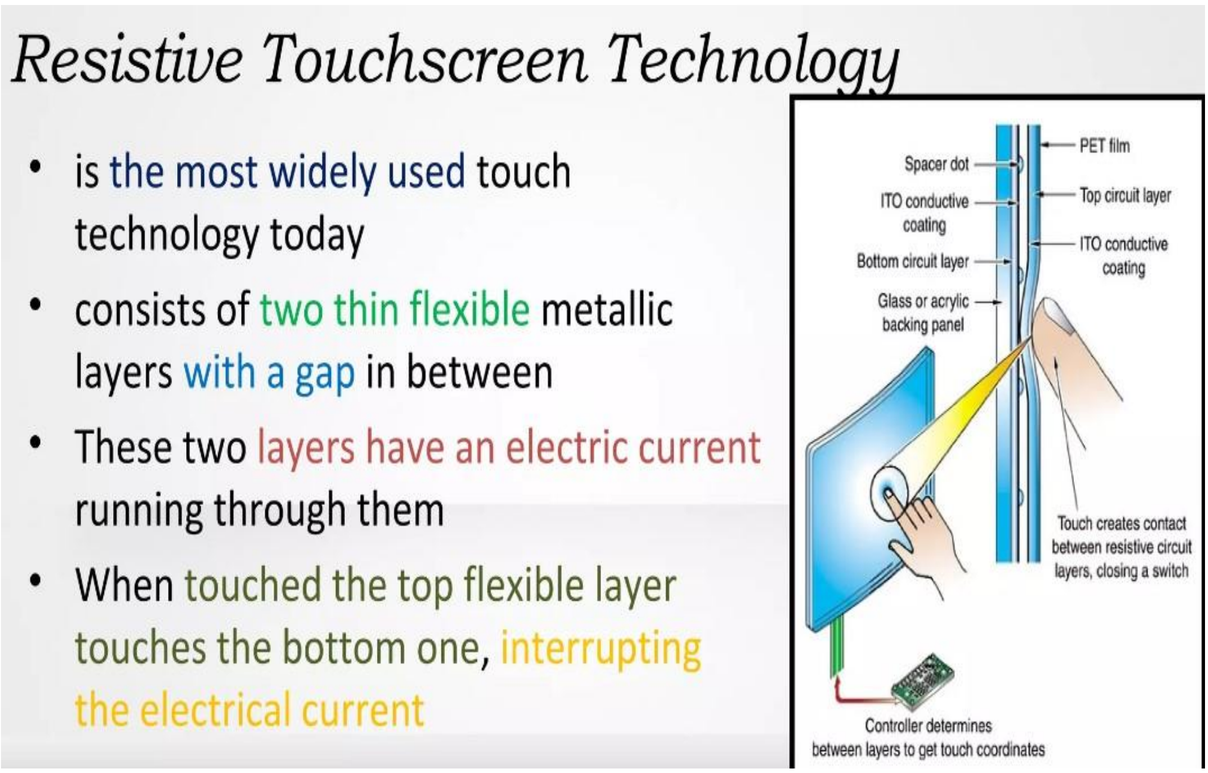


Touch Sensor Principle

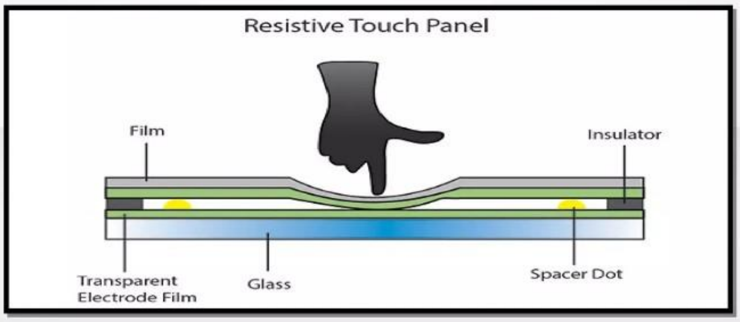
Types of Touch Sensors

The touch sensors are classified into two types. They are

a. Capacitive Touch Sensor
b. Resistive Touch Sensor



- The device notices this and detects the point of contact by the change in electrical flow
- responds to pressure and doesn't care what object applies the pressure
- Swiping and multi touch do not work, because this technology only registers one touch point ⟹ doesn't work on smartphones or tablets



Advantages

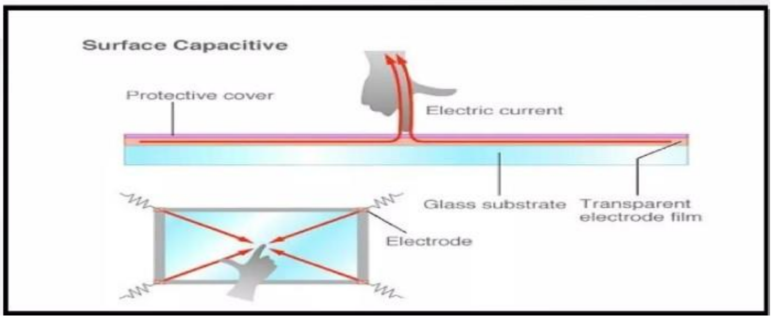
- Use pretty much any object to touch - finger, stylus, pen, gloved hand, etc.
- Solid feel
- Lowest cost
- Low power consumption
- Resistant to surface contaminants and liquids - dust, oil, grease, water, etc.

Disadvantages

- Image clarity not as great as other technologies
- Outer touch layer is vulnerable to damage - scratching, poking with sharp objects, etc.

Capacitive Touchscreen Technology

- uses a transparent electrode layer ⟹ is placed on top of a glass panel and covered by a protective layer
- When a finger touches the touchscreen ⟹ some of the electrical charge transfers from the screen to the user
- Sensors in all four corners of the screen detect the decrease of electric current ⟹ The controller then determines the touch point
- can only be activated when touched by human skin or a stylus holding an electrical charge



Advantages

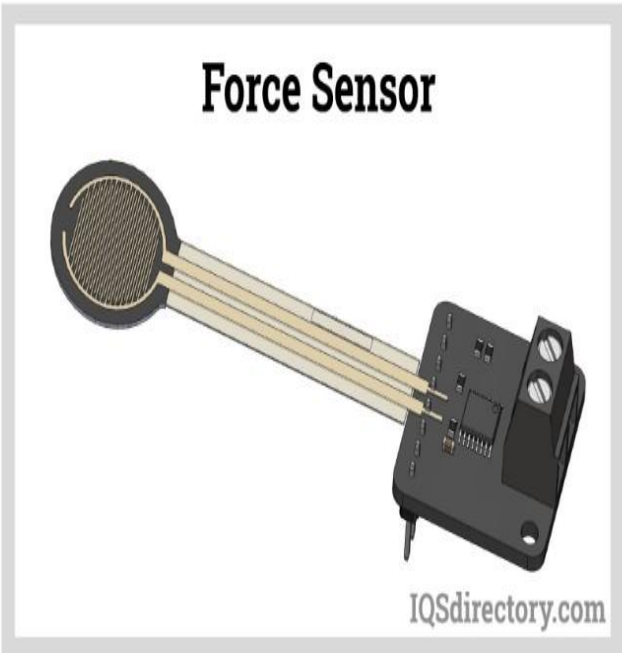
- Great image clarity (better than resistive touch)
- Durable screen
- Excellent resistance to surface contaminants and liquids - dust, grease and water
- High scratch resistance

Disadvantages

- Only works with bare finger or special capacitive stylus
- Sensitive to EMI (Electromagnetic Interference) and RFI (Radio Frequency Interference)

Force Sensor

Force sensors are transducers that transform mechanical input forces like weight, tension, compression, torque, strain, stress, or pressure into an electrical output signal whose value can be used to represent the force's magnitude. To notify operators or act as inputs for controlling machinery and processes, the signals may be delivered to indicators, controllers, or computers

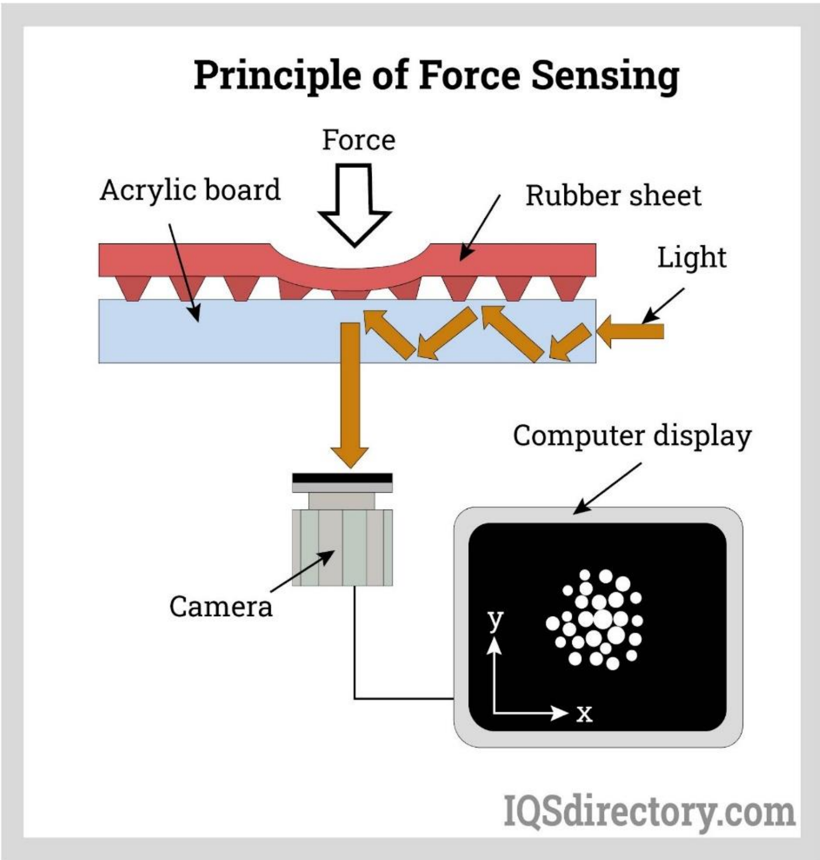


Force Sensor Principle of Operation

The fundamental idea behind how force sensors function is that they react to the applied force and transform its value into a quantifiable value. Based on diverse sensing components, there are numerous types of force sensors on the market. Certain substances are referred to as force-sensing resistors which can alter their resistance ratings when they are subjected to a force.

They typically include electrodes and a sensing material. The force imparted to an object can be measured with the use of a force sensor. The applied force can be determined by measuring how much the resistance values of force-sensing resistors vary. Force-Sensing Resistors are used to design the majority of force sensors.

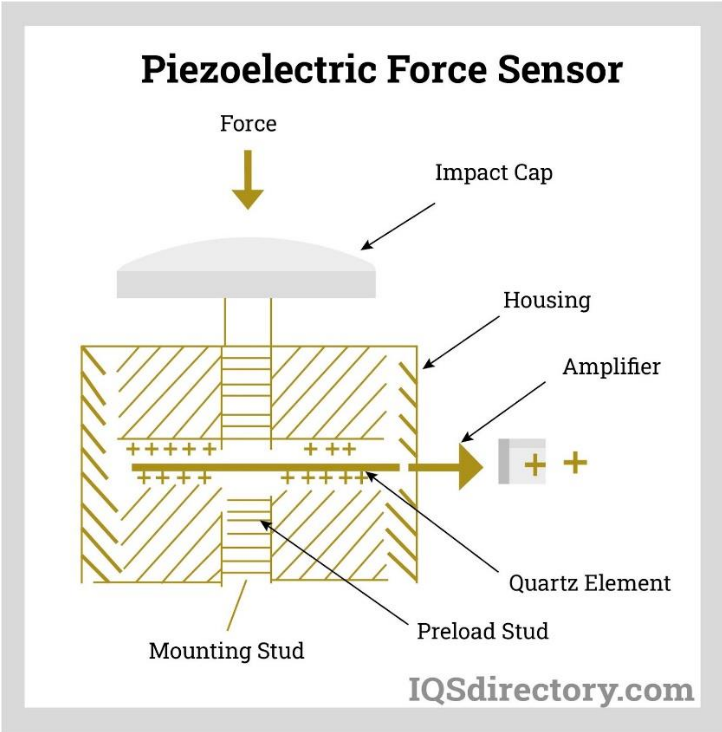
A force-sensing resistor's operation is based on the contact resistance property. In force-sensing resistors, a conductive polymer sheet changes resistance when force is predictably applied to its surface. This film is made up of a matrix of sub-micrometer-sized, electrically conducting, and non-conducting particles.



Types of Force Sensors

a. Piezoelectric Force Sensor

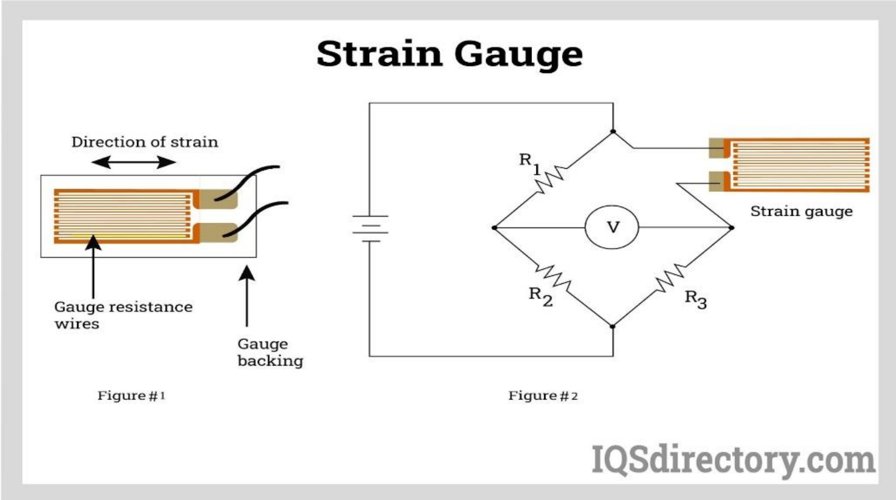
A gadget that utilizes the piezoelectric effect is an example of a piezoelectric force sensor. Some materials produce an electric voltage when mechanical stress or forces are applied along specific planes. By comparing the output of a calibrated piezoelectric force sensor with the output of a traditional foil strain gauge sensor and comparing the transfer functions of both sensors over a frequency range of 5-500 Hz.



b. Strain Gauge

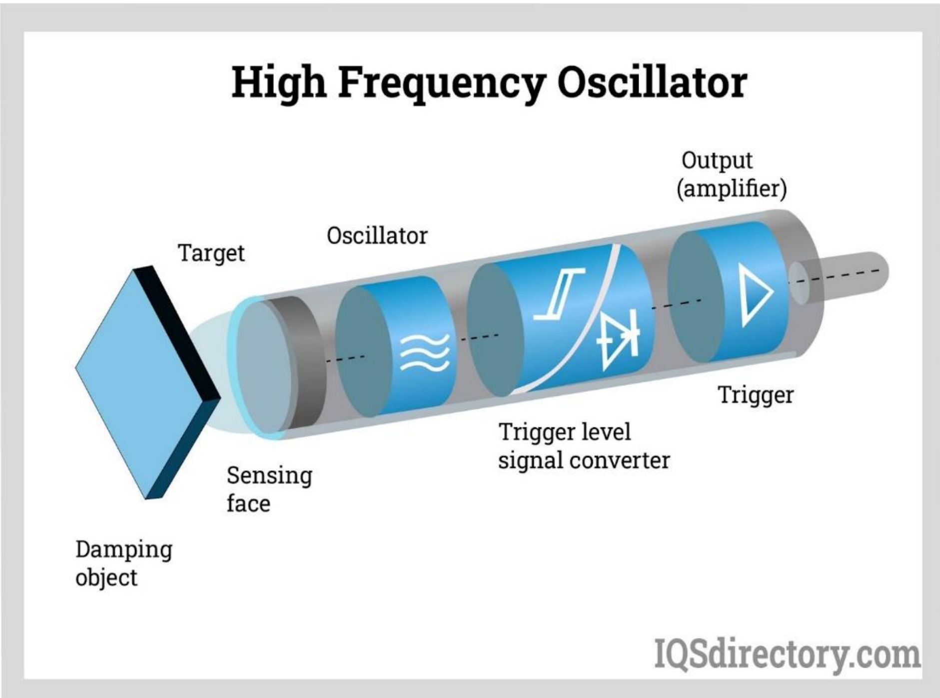
A strain gauge sensor uses elastic deformation of the strain gauge to convert an applied force to an electrical signal. Both static and dynamic forces may be present. Weight, acceleration, and pressure are a few examples. The strain gauge material's internal deformation alters the gauge's internal dimensions.

The wire lengthens and narrows when the force is parallel to the gauge's thin wiring and tensile, much like tugging a rubber band. As a result, the gauge's resistance rises. Similar to stretching, compressing reduces resistance by widening and shortening the wire. The amount of applied force directly relates to the change in the material's electrical resistance in either direction.



c. High-frequency Oscillation

An electromagnetic field with a high frequency is produced by the oscillator and is emitted from the switch's sensing face. Eddy currents are generated inside conductive metal when it comes into contact with this electromagnetic field, changing the oscillations' amplitude. The outcome is a voltage change at the oscillator's output, which modifies the trigger's state and the output state.



Applications of Force Sensor

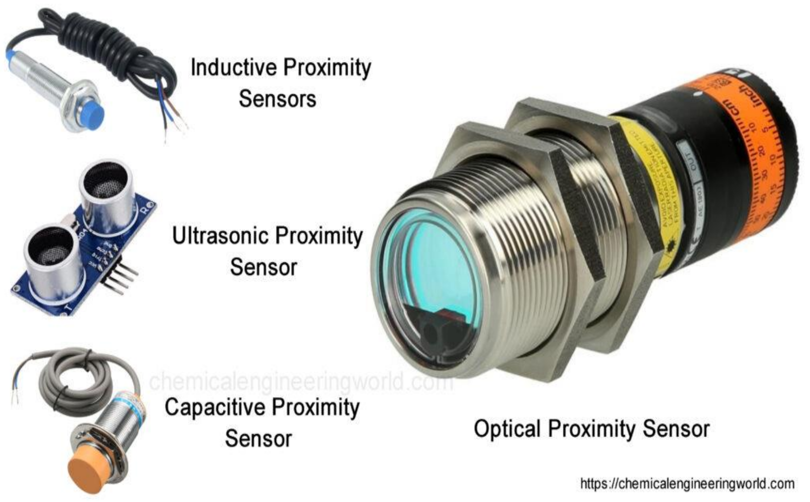
The main usage of the Force sensor is to measure the amount of force applied. There are various types and sizes of force sensors available for different types of applications. Some of the applications of Force sensor that uses force-sensing resistors includes pressure-sensing buttons, in musical instruments, as car-occupancy sensors, in artificial limbs, in foot-pronation systems, augmented reality etc.

Examples of Force Sensors

There are many types of force sensors available for different types of applications. Some of the examples of force sensors are Load cells, pneumatic load cells, Capacitive Load cells, Strain gauge load cells, hydraulic load cells, etc.

PROXIMITY SENSOR

- A non-contact sensor that is used to detect the presence of a target or object is called a proximity sensor.
- This sensor is capable of detecting close-by objects through electromagnetic radiation beams or electromagnetic fields without any physical contact.
- The detection of the target can be done based on the kind of proximity sensor by using light, sound, electromagnetic fields, IR, etc.
- This sensor is most frequently used in mobile phones, so the role of the proximity sensor in mobile is to detect a nearby person.



Working Principle

A non-contact sensor like a proximity sensor detects the existence of a target once it enters the field of the sensor. The detection of a target can be done based on the kind of sensor by using light, sound, IR otherwise electromagnetic fields.

These kinds of sensors are utilized in recycling plants, phones, anti-aircraft systems, assembly lines & self-driving cars. There are different kinds of proximity sensors available where each sensor detects targets in dissimilar methods.

Proximity Sensor Types

There are different kinds of proximity sensors used in different applications but each type of sensor detects an object in a different way they are inductive proximity, capacitive proximity, ultrasonic proximity, and optical proximity. But the most frequently used proximity sensors are inductive proximity & capacitive proximity.

1. Inductive Proximity Sensor

- These are contactless sensors used for detecting metal objects. This kind of sensor mainly depends on the induction law.
- Once a metallic object approaches it, then it drives an oscillator through a coil. This sensor includes two versions & includes four essential components which are discussed below.

The versions of inductive proximity sensors are unshielded and shielded.

a. Shielded Version

In the shielded version, the generated electromagnetic field is concentrated in the face, where the sensor coil sides are covered up.

b. Unshielded Version

- In the unshielded version, the generated electromagnetic field through the sensor coil is clear, so it allows for wider & greater detecting distances.
- Inductive proximity sensor includes four major components like an oscillator, coil, Schmitt trigger, & output switching circuit.
- When an AC is provided to the sensor coil, an electromagnetic detection field can be generated. Once a metal object is placed near the magnetic field, then eddy currents can be formed & result in coil inductance alters.

- Once coil inductance alters, then the circuit will change the output switch of the sensor which is monitoring continuously.

- Here whenever a target is not there, these sensors oscillate continuously and when an object is there then only the switch will trigger.
- The inductive proximity sensor is mainly used for industrial and security purposes like metal object detection, land mines, automation machine production, etc.

2. Capacitive Proximity Sensor

- These types of sensors are also contactless, used to notice both the objects like metallic & non-metallic that includes powders, liquid, granular, etc. This sensor works by noticing a change within capacitance.
- The main difference is it includes two charging plates like one internal & one external for capacitance. The connection of these two plates can be done like this; the internal plate is connected to the oscillator whereas the external plate is used as the detecting surface.
- This kind of proximity sensor generates an electrostatic field. Once an object reaches the detecting region, then both plate's capacitance also increases those results in amplitude gain of an oscillator.
- This gain activates the output switch of the sensor. These sensors oscillate only once the target is present.
- The applications of capacitive proximity sensors include Industrial processes, controlling moisture, touch applications, etc.

3. Ultrasonic Proximity Sensor

- These sensors are one kind of proximity sensor that is utilized in several applications of automation as well as manufacturing.
- The main purpose of this sensor is to detect objects and measurement of distance.
- These sensors are used in processing beverages, food & several packaging applications.

4. Magnetic Proximity Sensor

- The magnetic proximity sensor simply detects the magnetic field based on mechanical principles.
- They detect the presence of a magnetic target which is characterized through its magnetic field.
- Once it enters the range of the sensor detection, then it activates the switching procedure.

Features

- Contactless sensing allows this sensor to detect without touching the object, ensuring the object waits well-conditioned.
- These sensors are almost unaffected through outside colors of targets because it detects simply physical changes.
- These kinds of sensors are applicable for broad temperature range usage & damp conditions different from your usual optical detection.
- These sensors are used in android and IOS based mobiles. It includes simple infrared technology that activates and deactivates the display based on your usage. For instance, it deactivates your mobile display once a phone call is continuing such that you wouldn't start something accidentally while placing it close to your ears.
- Longer service life as compared to other sensors.
- Response rate speed is high.

APPLICATIONS

- The proximity sensor is mainly used to notice the existence of any target without touching.
- It detects as well as calculates any changes within the nearby location.
- These are utilized in mobile phones
- This kind of sensor is mainly used within the automation system of the home to switch on the light automatically once any person there within the room.
- These are used in machine tools, rolling mills & automation systems.
- This sensor is used to count the objects which are moving in the conveyor belt.
- This sensor is used for measurement of speed, direction rotation measurement of electrical motors & distance measurement of vehicles.
- The photoelectric proximity sensor is mainly used for plastic object detection.

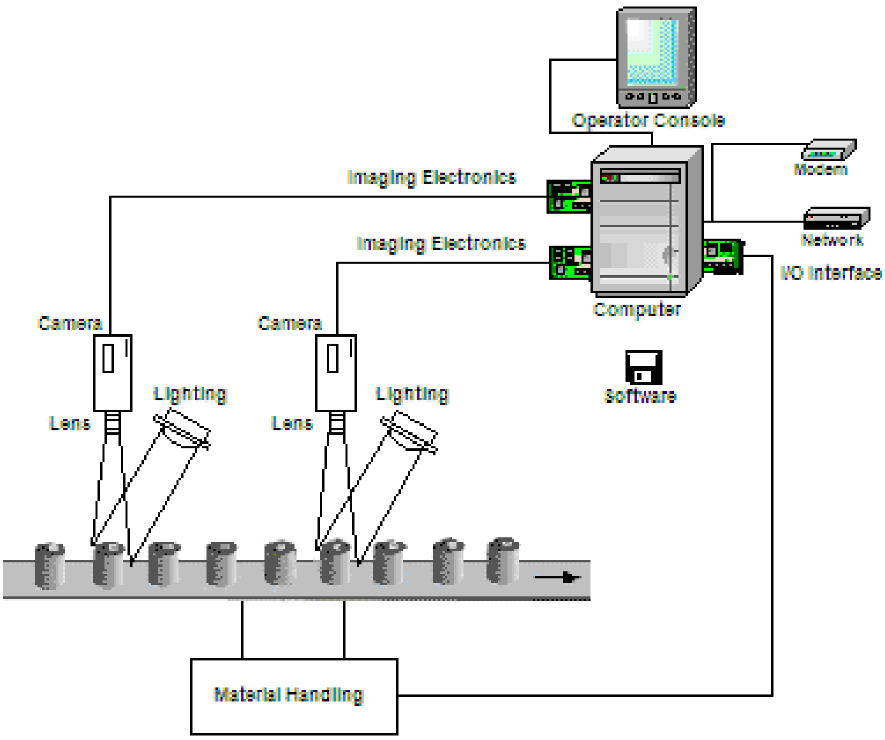
VISION SENSOR

- Vision sensors (a.k.a. machine vision or vision systems) are products consisting of a video camera, display and interface, and computer processor to automate industrial processes and decisions.
- These are commonly used for measurement, pass/fail decisions, and other observable characteristics relating to product quality.

- When the camera has an integral processor, it is called a smart camera.

Vision Sensor Operation

- Machine vision systems provide data interchange between a video camera and computer-based software.
- The images captured are compared against criteria specified by the operator to determine subsequent actions.
- This criteria can be based on barcodes, blot/stain detection, sizing and alignment, and a variety of other characteristics that can be determined solely by non-contact examination.
- Vision sensors are particularly helpful when multiple features must be inspected on the product. There is usually some degree of tolerance for each article.
- Vision sensors are considerably quicker and more accurate than humans employed for the same task.



Tasks of machine vision systems include, but are not limited to, the following:

- High-speed product inspection (quality control)
- Measurement
- Counting
- Sorting
- Locating
- Decoding
- Robot guidance

Multiple jobs can be performed simultaneously and are commonly achieved by the software's identification of one or more of the following when placed in the camera's field of view.

- Shape, pattern, or color
- Object detection, including object geometry and assembly
- Alignment
- Barcode or alphanumeric symbol recognition (OCR)

Vision Sensor Disadvantages

- While the benefits of machine vision sensors are substantial, the disadvantages of machine vision can be prohibitive.
- Machine vision systems are unable to cope with unforeseen circumstances and input.
- While ultimately cost-saving, high development costs can be expected for installation and personnel training.
- Constant levels of appropriate illumination can be difficult to maintain, and cameras can have difficulties isolating products in congested environments.

Applications

- The advantages of vision sensors are considerable, and many processes that involve a human inspection can utilize vision sensors to maximum efficiency.
- Industries already employing machine vision systems include food packaging and beverage bottling; automotive, electronics, and semiconductor assembly; and pharmaceutical companies.
- Common tasks for machine vision include robot guidance, pick-and-place processes, and counting.
- Railroad companies use machine vision for hi-rail automated rail inspection.

INTERNAL SENSORS

- Internal sensors are used to measure aspects about the robot itself, such as how fast it is going and at what angle it is facing.
- When we consider how fast a robot is going, we need to know about its speed, which is about how its position is changing, but also the direction of travel.

TYPES OF INTERNAL SENSORS

1. POSITION SENSORS

- Position sensor definition: It is a sensor that is used to detect the object's movement & convert it into signals which are suitable for transmission, control, or processing.
- These sensors are normally used for measuring the distance of the body from the reference position.
- So it detects how far the body has moved from one position to another and this output is frequently used as feedback to the control system to take appropriate action.
- The position sensor working mainly depends on providing motion control, counting & encoding different tasks through determining the existence or nonexistence of a target or otherwise by detecting its speed, direction, distance, or motion.
- The specific parameters of a position sensor will define its performance but they may change based on the type of sensor being selected.
- Some main specifications of position sensors include a range of measurement, accuracy, resolution, linearity, and repeatability**.**

Position Sensor Types

There are different types of position sensors available in the market which are discussed below.

*a.*Capacitive Displacement Sensor

- These sensors are non-contact devices, used for measuring any conductive target's high-resolution of the position & position change.
- These sensors are also used for measuring the density or thickness of non-conductive materials.
- Capacitive displacement sensors are used in different applications like semiconductor processing, precision equipment assembly like measurements of precision thickness, disk drives, assembly line testing & machine tool metrology.

b.Hall Effect Sensor

- A Hall Effect sensor is also known as a Hall sensor, which is used to detect the magnitude & presence of a magnetic field through the Hall Effect.
- These sensors are applicable in different applications like positioning, proximity sensing, detection of speed & current detecting.

c. Linear Variable Differential Transformer

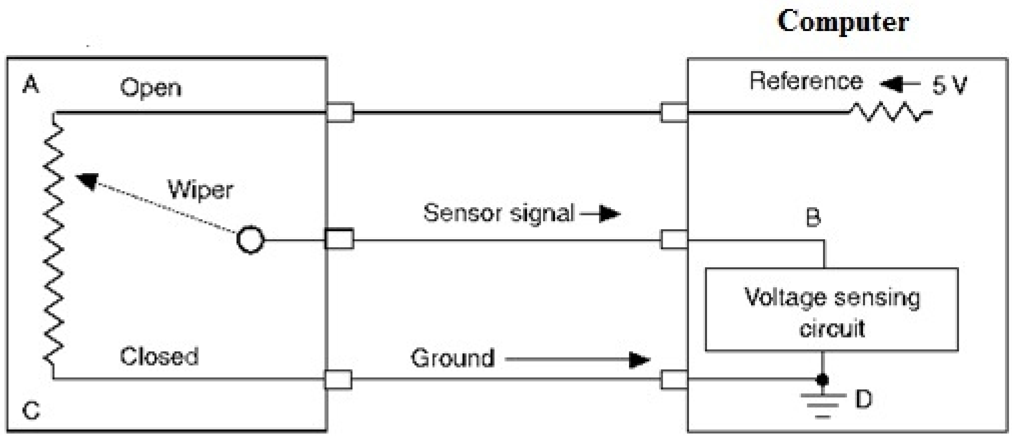
- The LVDT or linear variable differential transformer is the most frequently used electromechanical sensor.
- This sensor is mainly used for converting vibrations or mechanical motion, particularly rectilinear motion into electric signals.

d. Potentiometer

- Potentiometers are either rotary or linear position sensors that use thermoplastic damped wipers & a resistive track to measure displacement very effectively.
- Linear type potentiometers measure displacement linearly whereas rotary type measures rotating displacement.
- The construction of both the potentiometers is the same where each type includes a wiper & a track or conductive element.
- But the shape of the track is different in each type of potentiometer.
- In linear type, the track is straight whereas in rotary type it is circular. The wiper simply moves with the track to determine the displacement through the input voltage.

Position Sensor Circuit Diagram & Working

- A common type of position sensor that is used to monitor rotary or linear motion is known as a potentiometer.
- It is a voltage divider that provides a changeable DC voltage reading for the computer.
- These types of sensors are normally used to determine the position of the air conditioning unit door, a valve, a seat track, etc.
- A potentiometer sensor circuit is used to measure the sum of voltage drop to find out the position.



- Generally, the potentiometer includes a wire wound resistor including a movable wiper in the center.
- A stable voltage value like 5 volts is given to the 'A' terminal.
- If the wiper is connected close to the 'A' terminal, then the voltage drop will be low, which is denoted through a high voltage signal back to the computer throughout the 'B' terminal.
- When the wiper moves in the direction of the 'C' terminal, then the sensor voltage signal toward the 'B' terminal will be decreased.
- Here, the computer interprets various voltage values into various shaft positions.
- Once the wiper moves across the resistor, then the unit position can be tracked through the computer.
- Since voltage applied must flow throughout the whole resistance, then temperature & other factors do not make incorrect & false sensor signals toward the computer.

- A rheostat is not precise & its use is limited within computer systems.

Advantages

The **advantages of position sensors** include the following.

- Potentiometric sensors are Inexpensive & high accurate.
- LVDT or RVDT sensors provide high accuracy & Heavy-duty and are less sensitive to harsh environments.
- Optical sensors are highly accurate and have high resolution.
- Magnetic & Hall effect sensors are less sensitive to liquid and heavy-duty.
- The magneto strictive sensor is accurate especially for long lengths and heavy-duty.

Disadvantages

The **disadvantages of position sensors** include the following.

- Potentiometric sensors are sensitive to dust, extreme temperatures, high wear & tear.
- LVDT or RVDT is expensive, heavy & bulky.
- Optical type sensors are sensitive to dust, temperatures, and delicate.
- Magnetic & Hall effect sensors are responsive to impact, interrupted through electrical wires, magnetic materials & Hysteresis.

Applications

- These sensors are essential devices to measure rotary or linear positions.
- These sensors are used where measurement of movement is required.
- These sensors are used in different industries like Automotive, Hydraulics, Motorsport, medical, Mobile Vehicle, Aerospace, etc.

- Some of the other applications mainly include medical equipment, packaging machines, Drive-by-wire cars, Fly-by-wire aircraft systems, Injection moulding machines, etc.

2. VELOCITY SENSOR

A velocity or speed sensor measures consecutive position measurements at known intervals and computes the time rate of change in the position values.

Velocity Sensors:

1) Tachometers
2) LSV
3) Piezoelectric Sensors
4) Accelerometer Sensor

Tachometers:

A most important device that is used to provide velocity feedback is the tachometer.

It is also known as rpm gauge, and revolution counter.

A tachometer is employed in a motor to calculate the rotational speed of a shaft.

The output is displayed as RPM (revolution per minute) in an analog device.

The two common types of a tachometer are:

1) AC tachometer

2) DC tachometer

1) AC tachometer:

It possesses primary and secondary stators with fixed windings, and a rotor with *permanent magnet*.
If the rotor is stationary, a constant output voltage will be obtained.
If the rotor is moving, *proportional to the speed* of a rotor is induced.
This type of tachometer cannot provide information of direction with only one output winding.

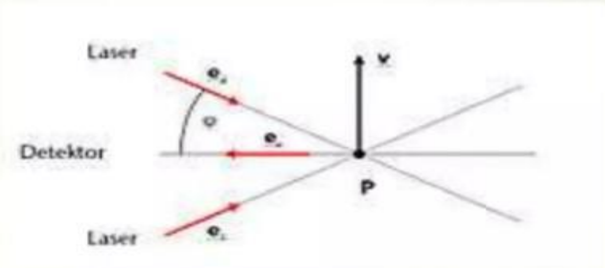
2) DC tachometer:

DC tachometer is the most commonly used instrument in the robotics.
It is a DC generator implemented to provide an output voltage that is proportional to the angular velocity of the armature.
In this mechanism, the rotor and rotational part will be attached directly. It has a stationary device called as commutator, which is connected with the split slip rings.

Laser surface velocimeter:

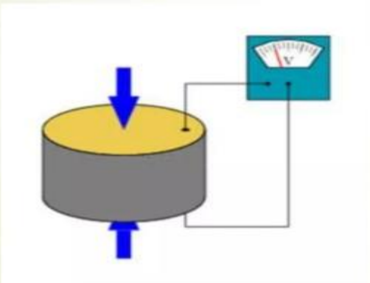
A laser surface velocimeter (LSV) is a non-contact optical speed sensor measuring velocity and length on moving surfaces. Laser surface velocimeters use the laser Doppler principle to evaluate the laser light scattered back from a moving object.

LSV is based on **Doppler effect** (or **Doppler** shift) that is the change in frequency of a wave (or other periodic event) for an observer moving relative to its source.



Piezoelectric Sensor:

A **piezoelectric sensor** is a device that uses the piezoelectric effect, to measure changes in velocity, pressure, acceleration, temperature, strain, or force by converting them to an electrical charge. The prefix piezo- is Greek for 'press'



Principle of operation:

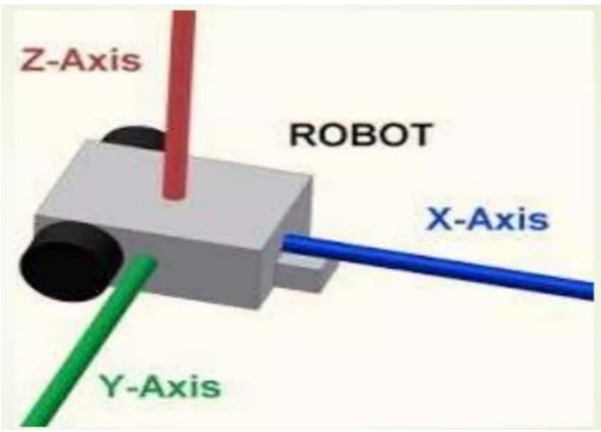
The way a piezoelectric material is cut produces three main operational modes:

1) Transverse
2) Longitudinal
3) Shear.

ACCELEROMETER SENSOR

An **accelerometer** is a device that measures proper acceleration(Time rate of change of velocity).

They are typically used in one of three modes:
1) As an inertial measurement of velocity and position;
2) As a sensor of inclination, tilt, or orientation in 2 or 3 dimensions, as referenced from the acceleration of gravity ($1\ g = 9.8 m/s^2$);
3) As a vibration or impact (shock) sensor



EXTERNAL SENSORS

- External is used to indicate that something is on the outside of a surface or body
- The external sensors, such as cameras, are used to capture user's trace, pose, salient body parts, related objects, scene context, etc.
- There are Two type of External sensors

  a. Contact Type
  b. Non-Contact Type

a. CONTACT TYPE

- Contact sensors make physical contact with the object that is being measured.
- An example of contact type sensors is – Temperature sensor

TEMPERATURE SENSOR

- Temperature sensors are devices that detect and measure coldness and heat and convert it into an electrical signal.
- Temperature sensors are utilized in our daily lives, be it in the form of domestic water heaters, thermometers, refrigerators, or microwaves.
- There is a wide range of applications of temperature sensors, including the geotechnical monitoring field.

What do Temperature Sensors do?

Temperature sensors are devices designed for measuring the degree of coolness and hotness in an object. The voltage across the diode determines the working of a temperature meter. The change of temperature varies directly proportional to the diode's resistance.

The cooler the temperature, the lesser the resistance will be and vice-versa. A measurement of the resistance across the diode is done, and the measurement is converted into units of temperature that are readable and displayed in numeric form over readout units. In the field of geotechnical monitoring, these temperature sensors are utilized in the measuring of internal temperatures of structures such as dams, bridges, power plants.

How Temperature Sensors Work

- The working principle of a temperature sensor is the voltage across the terminals of the diode.
- If there is an increase in the voltage, the temperature also increases.
- This is followed by a drop in the voltage between the terminals of the transistor of base and emitter in a diode.
- There are also temperature sensors that work on the principle of stress change caused by changes in temperature.
- In a vibrating wire temperature meter, dissimilar metals have different linear coefficients of expansion.
- It mainly consists of a magnetic stretched wire of high tensile strength with two ends fixed to any dissimilar metal so that any temperature change will directly affect the tension in the wire and its natural vibration frequency.
- The dissimilar metal can be made from aluminum since it has a larger linear expansion coefficient than steel.
- When the conversion of the temperature signal into frequency occurs, the very same read-out unit that is used for other vibrating wire sensors can also be utilized in the monitoring of temperature also.
- The specially built vibrating wire sensor is the one that senses the temperature change and then the temperature change is converted into an electrical signal which is then transmitted to the read out the unit as a frequency.

Types of Temperature Sensors

1. Resistance Temperature Detector (RTD)

- This is known as the resistance thermometer and uses the resistance of the RTD element with temperature to measure the temperature.
- Different types of materials can be used to make the metal.
- The materials include nickel, platinum, and copper. However, platinum is the most accurate and therefore the most expensive.

2. Thermistor Sensor

- This type of temperature sensor displays a change that is precise, predictable, and large in the alteration of different temperatures.
- With a change this large, it means that the reflection of the temperatures occurs rapidly and accurately.
- The NTC thermistor requires linearization because of the size and speed involved, so some mathematics are involved.

3. Thermometer Temperature Sensor

- This type of temperature sensor is the standard temperature sensor, particularly the mercury-filled glass tube.
- However, thermometers exist in several types.
- Glass thermometers can contain either ethanol or mercury, although nowadays the main liquid used in these thermometers is ethanol.

Benefits of Temperature Sensors

The benefits of temperature sensors include:

- Temperature sensors are precise, extremely reliable, and have a low cost.
- Temperature sensors are suitable for both embedded and surface applications.
- They provide low thermal mass resulting in a fast response time.
- The vibrating wire temperature sensor is completely interchangeable; all sensors can be read by one indicator.
- Temperature sensors are available with indicators for direct display of temperature.
- Temperature probes exhibit excellent hysteresis and linearity.
- The technology of the vibrating wire ensures long term stability, easy and quick readout.
- Temperature sensors perfectly suit remote scanning, reading, and data logging.

APPLICATIONS

- Medical Applications
- Uses in Motorsports
- Scientific and Laboratory applications
- Domestic Appliances

b. Non-Contact Sensor

- Non-contact sensors detect or measure a physical property without making direct contact with the target, the object that is being monitored or measured.
- They are used to measure physical properties such as thickness, proximity, displacement, or distance.
- An example of Non-contact type sensors is – **Ultrasonic Sensor**

ULTRASONIC SENSOR

The ultrasonic sensor is an electronic device used to measure distances. Because, measuring distance is an essential factor in many applications such as robotic control, vehicle detection etc. Sensors such as optical and sound are the most helpful.

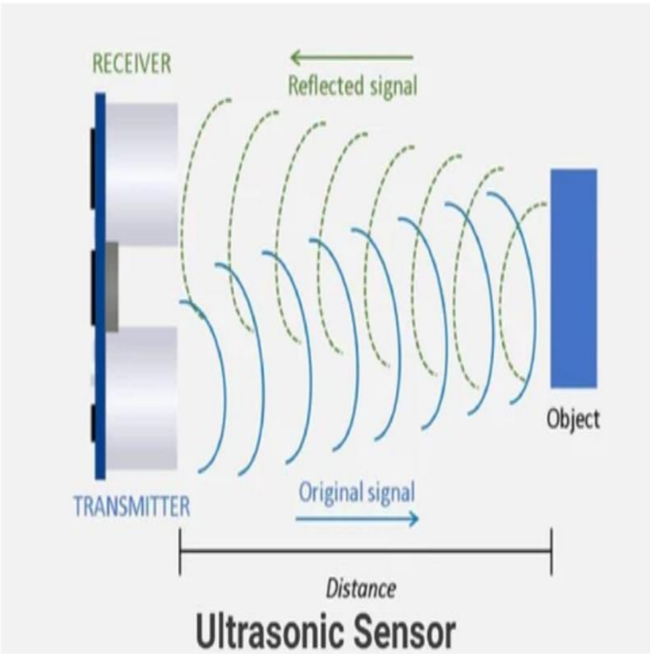
Ultrasonic sensors are used as proximity sensors. They can be found in parking technology and anti-collision safety systems. Ultrasonic sensors are also used in robotic obstacle detection systems and manufacturing engineering. Compared to infrared (IR) sensors in proximity sensing applications, ultrasonic sensors are less susceptible to interference from smoke, gases, and other airborne particles (although the physical component is subject to variables such as heat).

Ultrasonic sensors are also used as level sensors to detect, monitor, and control liquid levels in closed vessels (such as chemical plant drums). Most notably, ultrasound technology has enabled the medical industry to image internal organs, identify tumours, and ensure the health of babies in the womb.

PRINCIPLE OF ULTRASONIC SENSOR

The principle of ultrasonic rangefinders is to measure the time it takes
the signal sent by a transmitter and propagated back to the receiver. As the name implies ultrasonic sensor operates on ultrasonic frequencies. Frequencies beyond our hearing range are known as ultrasonic frequencies. Those frequencies are above 20k Hertz.

They are the all-rounders of sensor technology and can be used in any industrial application. There are several types of objects that can be detected, including solids, liquids, granules, and powders. They reliably detect transparent or glossy objects, as well as objects whose colors change.



HOW ULTRASONIC SENSOR WORKS?

An ultrasonic sensor is an electronic device that measures the distance to an object by emitting ultrasonic waves and converting the reflected sound into electrical signals. Ultrasound travels faster than audible sound (that is, sound that humans can hear). An ultrasonic sensor consists of two main components: a transmitter (which uses a piezoelectric crystal to emit sound) and a receiver.

While some sensors use separate sound emitters and receivers, it is also feasible to merge both functions into a single device by using an ultrasonic element to switch between sending and receiving signals in a continuous cycle. The transmitter of the module transmits an ultrasonic sound. This sound will be reflected if an object is present in front of the ultrasonic sensor. The reflected sound is received by the receiver present in the same module. An ultrasonic signal is propagated by a wave at an angle of 30°. The above-depicted Figure illustrates how the ultrasonic signal propagates from the transmitter. Measuring angles should be at least 15° for maximum accuracy. In this case, external objects that fall under this measurement angle interfere with determining the distance to the desired object.

The distance is determined by measuring the travel time of ultrasonic sound and its speed.

Distance = Time x Speed of sound / 2

APPLICATIONS

ULTRASONIC ANEMOMETERS:

Anemometers are often used in weather stations because they efficiently detect wind speed and direction. 2D anemometers can only measure the horizontal component of wind speed and direction, while 3D anemometers can also measure the vertical component of wind.

Ultrasonic anemometers can not only measure wind speed and direction, but also temperature. This is because the ultrasonic velocity is affected by temperature changes independently of pressure changes. Temperature is calculated by measuring the ultrasonic velocity change.

TIDE GAUGE:

It is used to monitor sea levels and detects tides, storm surges, tsunamis, swells and other coastal processes. Tide gauges can detect water levels in real-time using ultrasonic sensors. Meters are often linked to an online database where records are kept, and the system can trigger alarms if dangerous situations occur.

TANK LEVEL:

Measuring the liquid level in a tank is like a level meter. However, in this case, the fluid may be freshwater, corrosive chemicals, or flammable fluids. In contrast to optical sensors and float switches, ultrasonic sensors are not in contact with liquids, which means less corrosion.

WEB-GUIDING SYSTEMS:

The purpose is to make sure the material is placed correctly. If the material is misaligned, the system mechanically moves the material back into the machine path. Ultrasonic sensors are well suited for web guiding, as the process requires non-contact, fast and efficient functionality.

CHARGE COUPLED DEVICE (CCD CAMERA)

- A CCD camera is a video camera that contains a charged-coupled device (CCD), which is a transistorized light sensor on an integrated circuit.
- In cameras, CCD enables them to take in visual information and convert it into an image or video.
- This allows for the use of cameras in access control systems because images no longer need to be captured on film to be visible.

How does a CCD camera work?

In terms of the working principle of CCD cameras, these video cameras capture an image and transfer it to the camera's memory system to record it as electronic data. CCD cameras' main accomplishment is the production of quality images without any distortion. Basically, the camera turns light into electricity. A CCD camera forms light sensitive elements called pixels which sit next to each other and form a particular image. CCD cameras have been in production for a long period of time and tend to have high quality pixels that produce a higher quality, low-noise image than any other camera.

Complementary Metal Oxide Semiconductor (CMOS CAMERA)

- A CMOS sensor or complementary metal-oxide-semiconductor sensor is one kind of electronic chip, used to change the photons into electrons for digital processing.
- These sensors are mainly used for creating images within digital cameras, digital CCTV cameras & digital video cameras.
- The CMOS sensor works on the principle of the photoelectric effect to change the photons into electrical energy.

How does CMOS Sensor Work?

- The image sensor within a camera system receives incident light that is focused through a lens. Based on the type of image sensors like CMOS or CCD, the camera system will transmit data to the next phase like either a digital signal or a voltage.

- CMOS sensors convert photons into electrons to a voltage & after that into a digital value through an on-chip ADC (Analog to Digital Converter).

- Depending on the manufacturer of the digital camera, the components used in the camera system and its design will be changed.
- The main function of this design is to change light into a digital signal then it can be examined to activate some further enhancement or user defined actions.
- The cameras which are used at the consumer level are available with additional components to store the image in memory, LCD & switches, and control knobs but machine vision type cameras do not include.

Difference between CMOS and CCD Sensors

CMOS Sensor	CCD Sensor
CMOS sensor is a metal oxide semiconductor chip, used to change light into an electrical signal.	It is a charge-coupled device, used to transmit electrically charged signals.
CMOS sensors are available in two types active pixel and passive pixel.	CCD sensors are available in three types like Full-Frame, Frame-Transfer & Interline-Transfer.
Low power consumption	Moderate to high power consumption
Moderate complexity	Low complexity
CMOS resolution is low to high	CCD resolution is low to high
Low uniformity	High uniformity
It has a moderate dynamic range	It has a low dynamic range
Moderate to the high noise level	Low noise level
The fill factor is moderate	The fill factor is high
Chip signal is digital	Chip signal is analog

Advantages

The **advantages of CMOS sensors** include the following.

- Power consumption is low.
- Less cost.
- Inferior dark noise will give a higher reliability image.
- The camera size as small as the readout logic can be included in a similar chip.
- Flexible readout because of the direct individual pixels addressing which allows more storage bin & limited scan possibilities.
- Higher sensitivity in the NIR range.

Disadvantages

The disadvantages of the CMOS sensor include the following.

- These sensors are more susceptible to noise and images are grainy sometimes.
- These sensors utilize more light for an enhanced image.
- In this sensor, every pixel executes its conversion.
- The homogeneity & quality of the image is low.

Applications

The applications of CMOS sensors include the following.

- These sensors are used in different fields like marine, automotive, manufacturing, aviation, healthcare, astronomy, medical & digital photography.
- CMOS sensors cover from automation to traffic-based applications like blind guidance, aiming systems, active or passive range finders, etc.
- These sensors change photons to electrons for use in digital processing.
- These sensors will create images within digital cameras, digital CCTV cameras & digital video cameras.
- These sensors are used in high-resolution cameras.